

Q.1 Explain Inheritance in Java and why multiple inheritance with classes is disallowed.

Ans: Inheritance is an Object-Oriented mechanism where a child class derives attributes and methods from a parent class using the `extends` keyword. It fosters code reuse and method overriding.

Java supports Single, Multilevel, and Hierarchical inheritance. However, multiple inheritance with classes is forbidden in Java to prevent the Diamond Problem, which causes ambiguity in method resolution.

Instead, Java supports multiple inheritance through Interfaces.

Q.2 Differentiate Method Overloading and Method Overriding in Java.

Ans: Method Overloading occurs within the same class where methods share the same name but have different parameter lists (type, count, or sequence). It is compile-time polymorphism.

Method Overriding occurs between subclass and superclass where the method signature is identical. It is runtime polymorphism resolved via dynamic method dispatch using the virtual method table (vtable).

Q.3 Describe JVM architecture and its runtime memory areas.

Ans: JVM runtime data areas consist of:

1. Method Area: stores class-level metadata and static variables.
2. Heap Area: stores all instantiated objects and instance variables.
3. JVM Stack: stores frames, local variables, and partial results.
4. PC Registers: keeps address of currently executing JVM instruction.
5. Native Method Stack: supports native C/C++ libraries.

Garbage Collection automatically reclaims unreferenced heap memory.

Q.4 Explain Exception Handling hierarchy and checked vs unchecked exceptions.

Ans: `Throwable` is the superclass of `Exception` and `Error`.

Checked exceptions inherit from `Exception` (e.g. `IOException`, `SQLException`) and must be caught or declared.

Unchecked exceptions inherit from `RuntimeException` (e.g. `NullPointerException`, `ArithmeticException`).

The `finally` block always executes regardless of exception, even with `return` in `try/catch`.

Q.5 Compare `ArrayList` vs `LinkedList` and `HashMap` vs `ConcurrentHashMap`.

Ans: `ArrayList` uses a resizable array with  $O(1)$  random access.

`LinkedList` uses doubly linked nodes with  $O(1)$  insertion at ends.

`HashMap` is not synchronized. `ConcurrentHashMap` provides thread safety using bucket-level locks.